



## MECHANICS OF FLIGHT I 25/26-S2

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### *Lab 2: Aircraft Performance - Group 28*

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## Introduction

This document reports on the second laboratory session on Flight Mechanics. The lab centers around aircraft performance across three different flight phases: Cruise, Motion in a vertical plane, and Turning maneuvers. The objective is to understand the control laws that govern them, as well as the dynamics of aircraft by integrating the equations of motion numerically and comparing with the analytical models derived in the theory sessions by use of  $\hat{V} - \hat{T}$  plots.

## 1 Cruise Performance Analysis

The following section tackles a horizontal level flight at  $H = 10,000$  m using a fixed thrust setting of  $\Pi = 0.8$ . A simulation is set up over 60 minutes of flight starting from 3 different initial conditions function of the base velocity ( $1.2V_{Em}$ ,  $0.6V_{Em}$ ,  $0.4V_{Em}$ ) in order to study the stability of the equilibrium points.

### 1.1 Temporal Evolution of the State Variables

The governing equations of motion were integrated numerically by use of the provided function and *MATLAB*'s `ode45`. Figure 1 shows the evolution of the evolution of the velocity, range, height and the weight.

The first check for a successful simulation is the fact that the altitude has indeed remained constant at 10 km, meaning cruise conditions have been satisfied. The linear decrease in weight is expected due to a constant consumption of fuel. This last factor induces a small change in both base velocity  $V_{Em}$ , and thrust  $T_{Em}$ , over time.

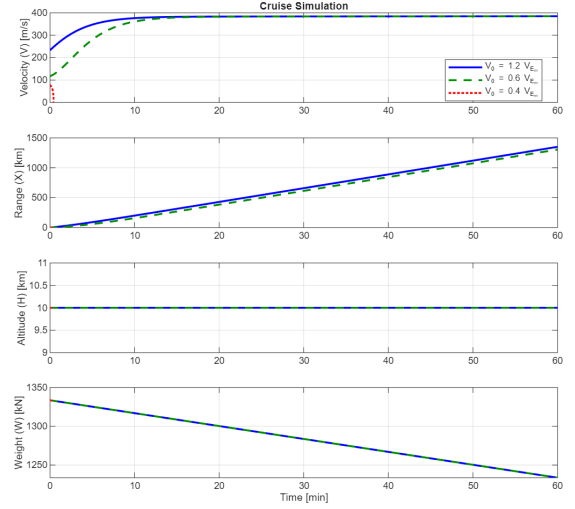


Figure 1: Evolution of state variables: Velocity ( $V$ ), Range ( $X$ ), Altitude ( $H$ ), and Weight ( $W$ ) over 60 minutes.

### 1.2 Dimensionless Trajectory vs. Analytical Polar

For the following section, both dimensionless variables  $\hat{v}$  and  $\hat{T}$  were converted from the aforementioned simulation in time, with the help of the provided functions `ISA` and `From_D_2_ND` in order to translate into the non-dimensional equivalent. The analytical solution provided by Eq.8:

$$\hat{T} = \frac{1}{2} \left( \hat{v}^2 + \frac{1}{\hat{v}^2} \right) \quad (1.1)$$

is represented alongside the results for the different starting velocities in Figure 2.

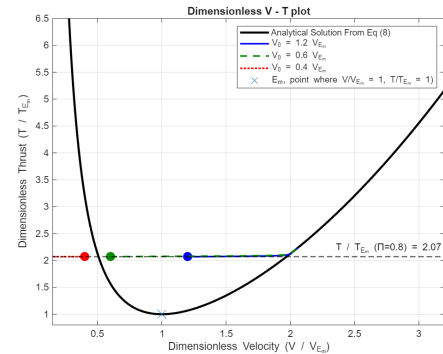


Figure 2: Dimensionless trajectory in the  $\hat{v} - \hat{T}$  plane compared with the analytical cruise polar.

The trajectory of the aircraft is mostly horizontal because the thrust remains almost constant (as mentioned before, there is a slight variation with weight), meaning there is little change in the vertical axis. Major adjustments to reach equilibrium are achieved through the velocity instead.

### 1.3 Results for the Different $v_0$

The following three initial cases help in discussing the stability of the flight regimes:

- **Case  $V_0 = 1.2V_{Em}$ :** We will call this starting point the "first regime", and it is stable. Since the available thrust is greater than the required thrust at this speed, the aircraft converges smoothly to the equilibrium point of high-speed, as can be seen in figure 2.
- **Case  $V_0 = 0.6V_{Em}$ :** We will call this starting point the "second regime", and it is situated in the backside of the power curve, or what is essentially left of  $\hat{V} = 1$ . This region is theoretically unstable, meaning that the pilot must avidly engage the controls of the aircraft for a smooth flight. In this case, the thrust setting of  $\Pi = 0.8$  is high enough to overcome induced drag, allowing the aircraft to accelerate, cross the value  $V = V_{Em}$  and thus stabilize into the "first regime".
- **Case  $V_0 = 0.4V_{Em}$ :** The initial velocity in this case is too low, falling behind the stall line, resulting in the velocity reaching 0, the flight becoming unsustainable, and a crash to occur as consequence. This happens as the induced drag -  $D \propto 1/V^2$  - exceeds the maximum thrust provided by  $\Pi = 0.8$ .

### 1.4 MATLAB vs. Simulink

Once the analysis performed on MATLAB is modeled in Simulink, we compare both methods for a problem such as the one given in this section of the lab:

#### MATLAB:

- *Advantages:* It is very efficient for automating tasks and let's us provide very personalized figures for ease of analysis. For example, for the 3 different starting velocities we used a for loop, and for the dimensionless plot we chose a way of displaying starting point and trajectory in a very precise manner.
- *Disadvantages:* Being just code, it is quite abstract, and necessitates either comments or proper sectioning to understand real physical phenomena.

#### Simulink:

- *Advantages:* Block diagrams better visually represent the aircraft's physics. The feedback loops make it more intuitive to see, for example, how the weight affects the required lift/drag throughout the simulation time. It also allows for greater visual interactivity between the functions as blocks, making it overall more engaging.
- *Disadvantages:* Changing the parameters for different cases is sluggish, and plots lack the custom factor that MATLAB plots provide. For example, the plot below represents the velocity in the third case, where the aircraft stalls early on. As the model does not allow for a side-by-side comparison of all 3 cases, it proves less intuitive to discern that only the first few seconds of this plot are useful, and that the rest of the simulation

is for after the plane has crashed. Without sufficient physical understanding of the problem, it could lead to wrong conclusions, like the velocity evolving through the remaining time as it does in the figure (results not valid as main assumptions violated after stall event).

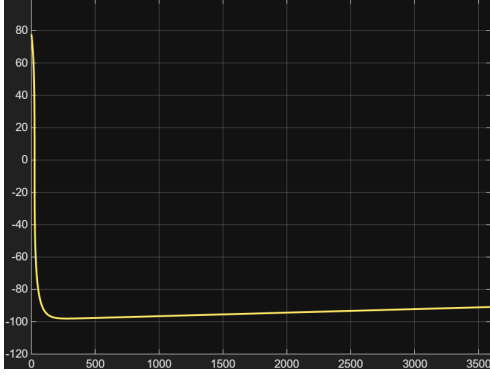


Figure 3: Range for Case 3, stall, as obtained through Simulink.

## 2 Motion in a Vertical Plane

The section that follows analyzes a descent maneuver at  $H = 15,000$  m for a constant elevation angle. Isolines for  $\gamma$  characterize the equilibrium conditions, and will be represented alongside the results of the analysis that is to be conducted.

### 2.1 Gamma Isolines

To obtain isolines for  $\gamma$  we employ MATLAB's *contour* for various values of it, as represented in Figure 4. Every curve represents a quasi-steady equilibrium state where longitudinal acceleration is null.

### 2.2 Minimum Thrust Setting for Specific Maneuver

To obtain the thrust level  $\Pi$  for conditions  $v = 1.2v_{Em}$ ,  $\gamma = -2^\circ$ , an altitude of 15 km and  $W = MTOW$  (as required to maintain a

steady descent). From the equilibrium equation  $T = D + W \sin \gamma$  and normalizing by using base thrust:

$$\hat{T}_{req} = \frac{1}{2} \left( \hat{v}^2 + \frac{\cos^2 \gamma}{\hat{v}^2} \right) + E_m \sin \gamma \quad (2.1)$$

The required thrust can then be compared to the base value at the given height to obtain that  $\Pi \approx 0.3204$ . This value sits close to a third of the available thrust, meaning the power plant must be throttled down significantly to maintain this constant descent.

### 2.3 Trajectory Analysis

After generating an initial condition vector, the equations of motion were numerically integrated at fixed thrust  $\Pi = 0.5$  and  $\gamma = -2^\circ$  - equating the lift to a weight component as  $L = W \cos \gamma$ . The result is superimposed to the  $\gamma$ -isolines in Figure 4.

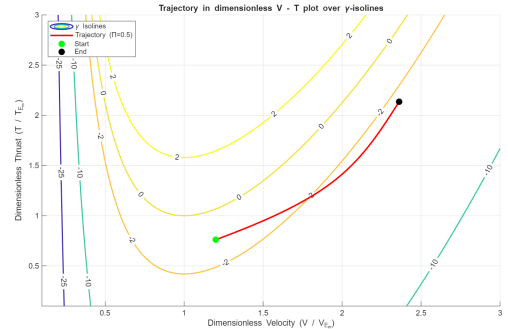


Figure 4:  $\gamma$ -isolines and numerical trajectory for  $\Pi = 0.5$  and  $\gamma = -2^\circ$ .

As can be seen, the trajectory does not exactly follow the  $\gamma = -2^\circ$  isoline. This is because the isoline corresponds with the thrust setting required for equilibrium, previously found to be  $\Pi \approx 0.38$ . However, since the selected setting is greater than this ( $\Pi = 0.5$ ) the aircraft experiences a positive acceleration. The evolution of the trajectory is also affected by a shift in the dimensionless quantities, forcing it to cross mul-

multiple states in a dynamic evolution. This shift in dimensionless quantities is driven solely by the change in altitude, which changes the density, and in turn the base quantities.

## 2.4 Simulink Model Analysis

Simulink in this case especially helps in visualizing the feedback loop for the variation of the altitude with density. The results for both  $V_0 = 0.5V_{Em}$  and  $V_0 = 1.2V_{Em}$  can be analyzed in the following figures:

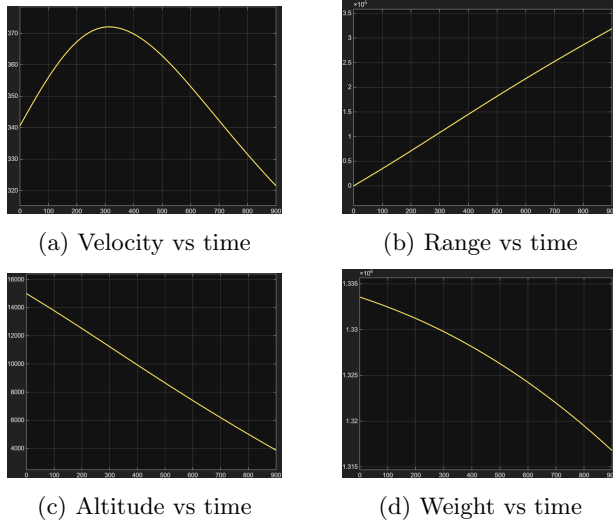


Figure 5: Simulink results for  $v_0 = 1.2V_{Em}$ ,  $\Pi = 0.5$ ,  $\gamma = -2$ .

Where we can see that for the first case every state variable behaves as expected. The aircraft is initially accelerated ( $\Pi = 0.5 > 0.32$ ), with peak velocity at  $t \approx 200s$ , and a reduction due to the domination of drag.

In the second case, we can see that in the same way as in the Cruise setup, at  $t \approx 150s$  deceleration happens until a crash occurs. Hence, the remaining time in the simulation is not only nonphysical, but also useless. This occurs because the model simply counts with no answer for a case where stall occurs, meaning the model is only valid above the minimum controllable speed, and  $v_0 = 0.5V_{Em}$  lies too deep in the

unstable region to recover.

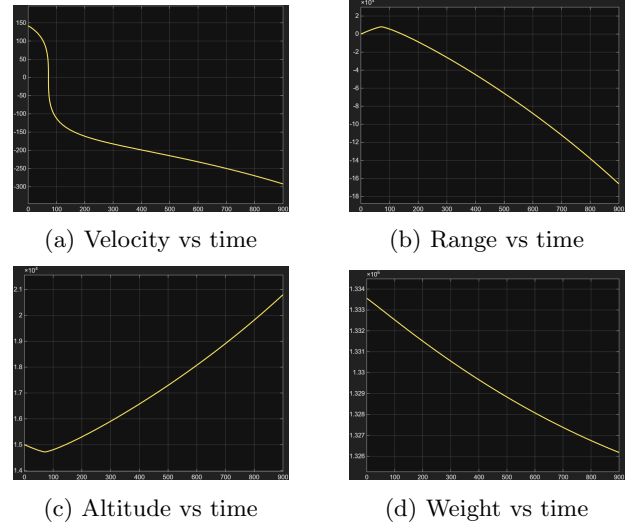


Figure 6: Simulink results for  $v_0 = 0.5V_{Em}$ ,  $\Pi = 0.5$ ,  $\gamma = -2$ .

## 3 Turning Maneuver Analysis

The following section centers on the performance of a horizontal, steady turn for an altitude of  $H = 10,000m$ .

### 3.1 Identification of State and Control Vectors

The vectors that define the turning maneuver are both the state and the control vectors, defined as:

- **State Vector:**  $x_s = [V, \chi, X, Y, W]^T$ , composed of velocity, heading angle, horizontal coordinates, and weight.
- **Control Vector:**  $x_c = [\Pi, C_L, \mu]^T$ , composed of thrust level, lift coefficient, and bank angle.

### 3.2 Isolines Map

Figure 7 shows the performance map for the requested variables: bank angle ( $\mu$ ), load factor

( $n$ ), dimensionless radius ( $Rg/V_{Em}^2$ ), and dimensionless turn rate ( $V_{Em}\dot{\chi}/g$ ). The isolines define the requirements for any state condition, and serve as map for obtaining every one of them depending on the velocity and thrust at  $H = 10,000\text{m}$  in the same way as an aviation chart would for the purposes of performing the maneuver at hand.

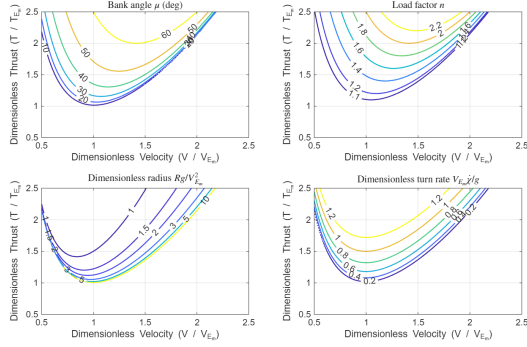


Figure 7: Turning performance isolines:  $\mu$ ,  $n$ , dimensionless radius, and turn rate.

### 3.3 Numerical Integration and Resulting Spiral Trajectory

Integration of the governing equations was performed for a duration of 5 hours. The starting conditions employed were  $X_0 = 0, Y_0 = 0, \chi = 0$ , whilst starting from  $\hat{v} = 1.1$  and  $\hat{T} = 1.2$ , with  $\Pi$  and  $\mu$  kept constant.

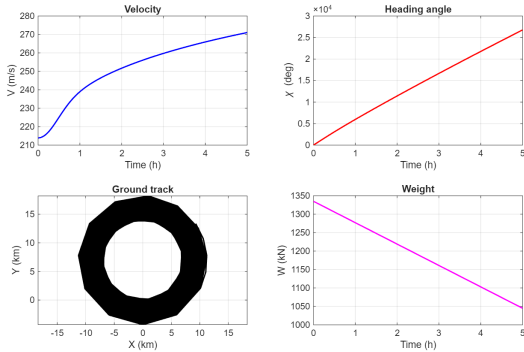


Figure 8: Temporal evolution of  $V$ ,  $\chi$ ,  $X$  vs  $Y$ , and  $W$  during a 5-hour turn.

In the resulting plots one can appreciate a steady increase in velocity, mainly due to the surplus

of thrust that is induced by the decrease in weight. This increase in velocity is seen reflected in the ground track that the aircraft follows, as  $R = V^2/(g \tan \mu)$  the radius increases, leading to a spiral to be described (in Figure 8 it is difficult to discern the separation between the lines since the increase in radius is fairly slow, but it is nonetheless a spiral).

### 3.4 Trajectory over $\mu$ -Isolines

The trajectory of the numerical simulation may be computed against the  $\mu$ -isolines for further analysis.

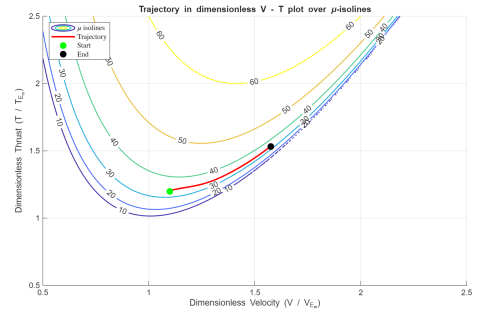


Figure 9: Numerical trajectory across  $\mu$  isolines.

The bank angle is constant, yet the dimensionless state ( $\hat{v}, \hat{T}$ ) drifts. This happens since both reference values decrease with burnt fuel as  $V_{Em} \propto \sqrt{W}$  and  $T_{Em} \propto W$ . This in turn makes both  $\hat{T}$  and  $\hat{v}$  increase, and there is a translation in the performance map toward regions of higher load factors and tighter dimensionless requirements.